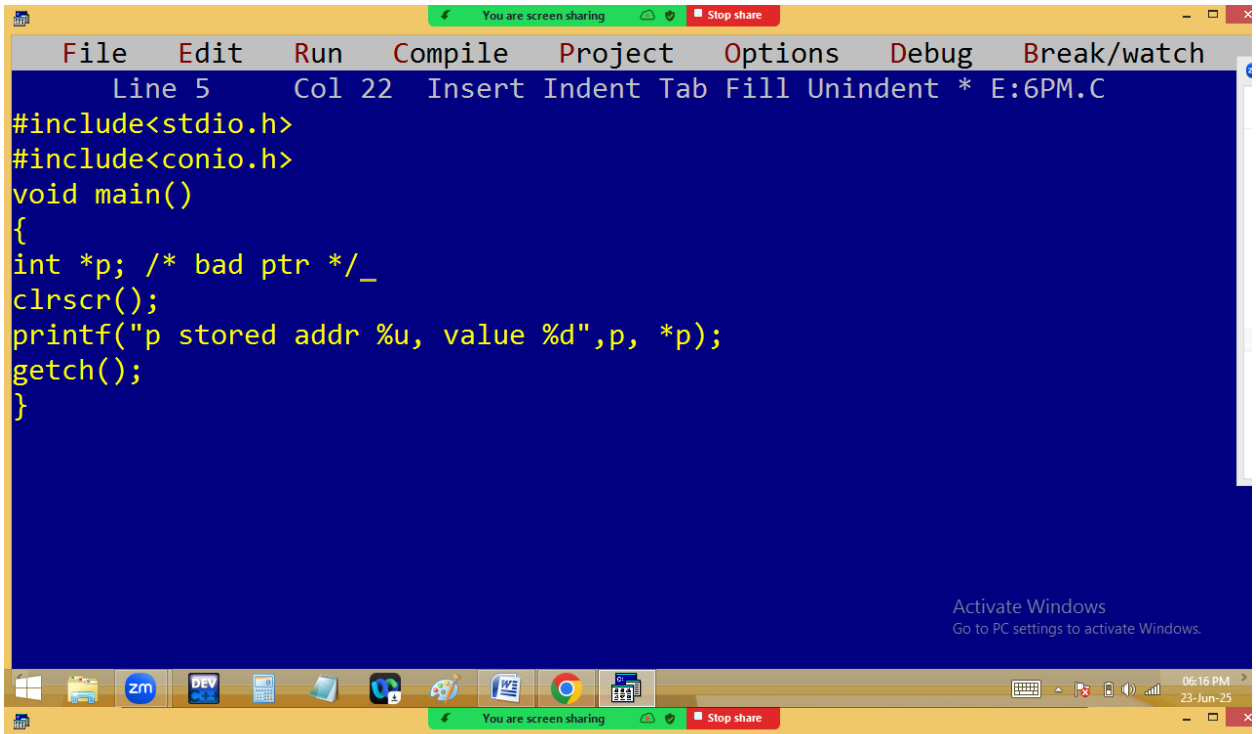
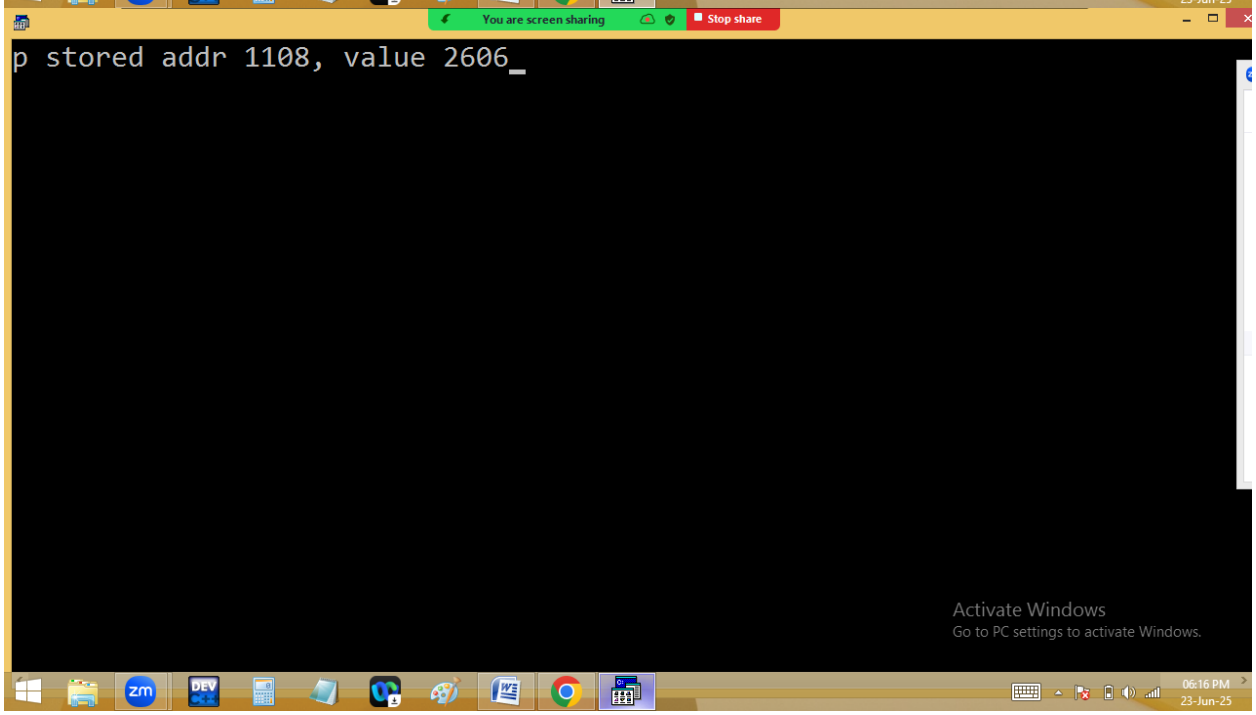
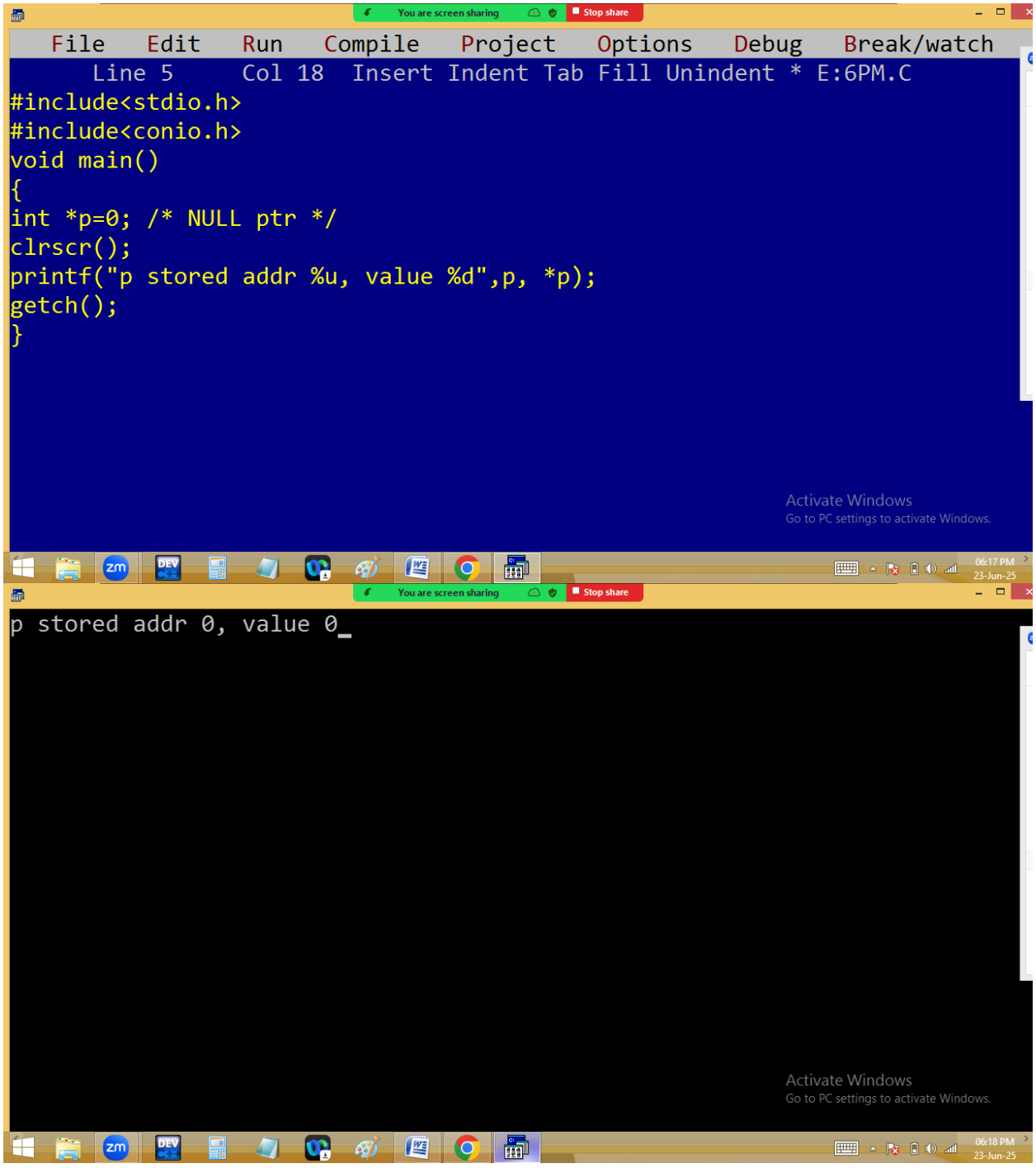
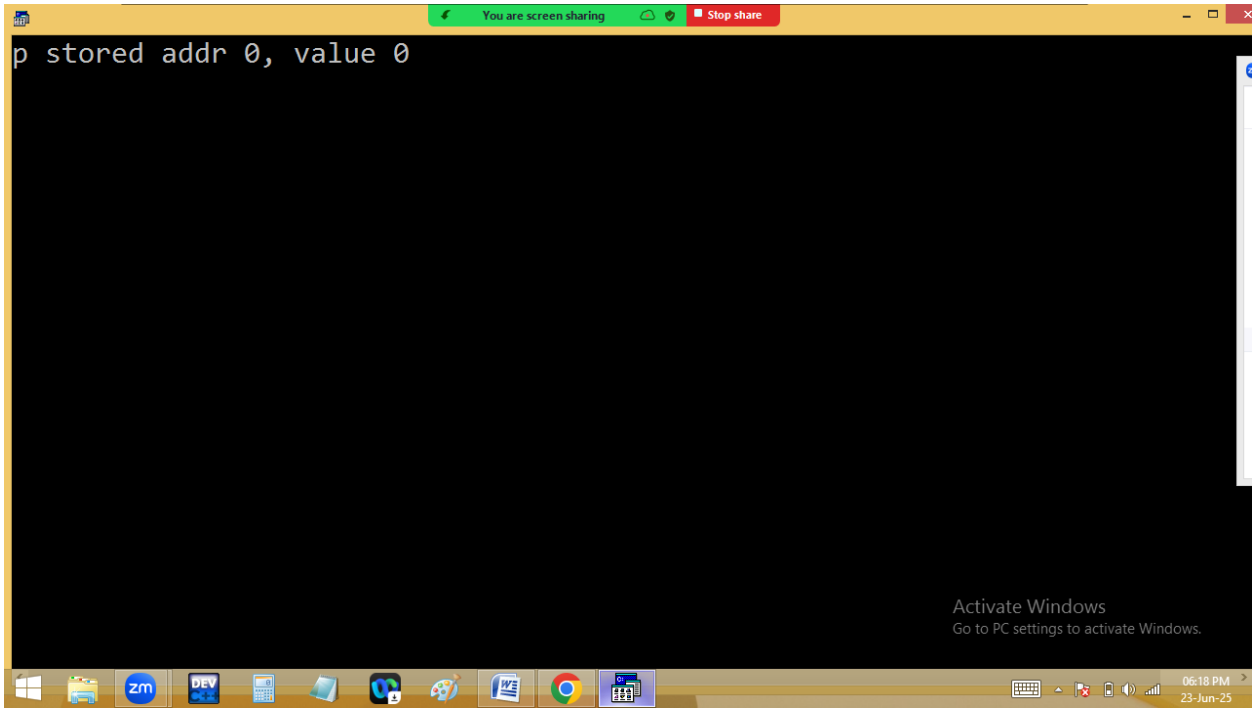
Bad / wild pointer: **A pointer is declared but not initialized. In this situation the pointer is storing some unknown address and value. This kind of pointer is called bad / wild pointer.**



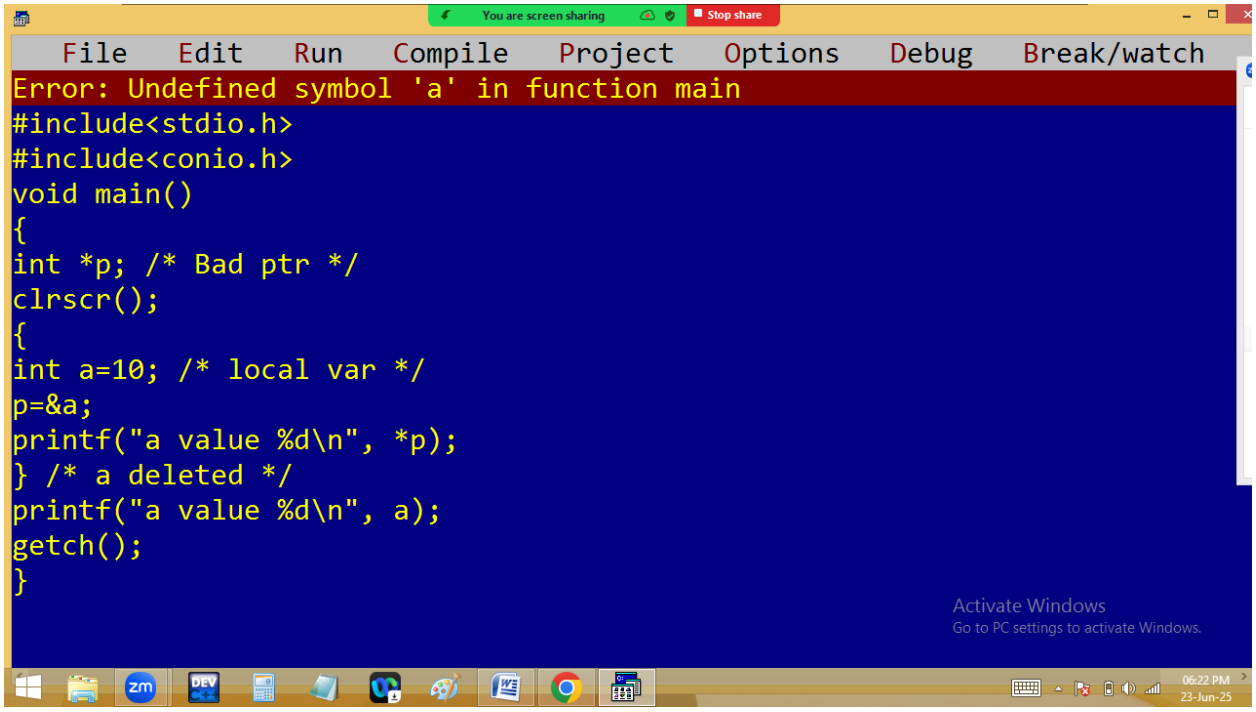


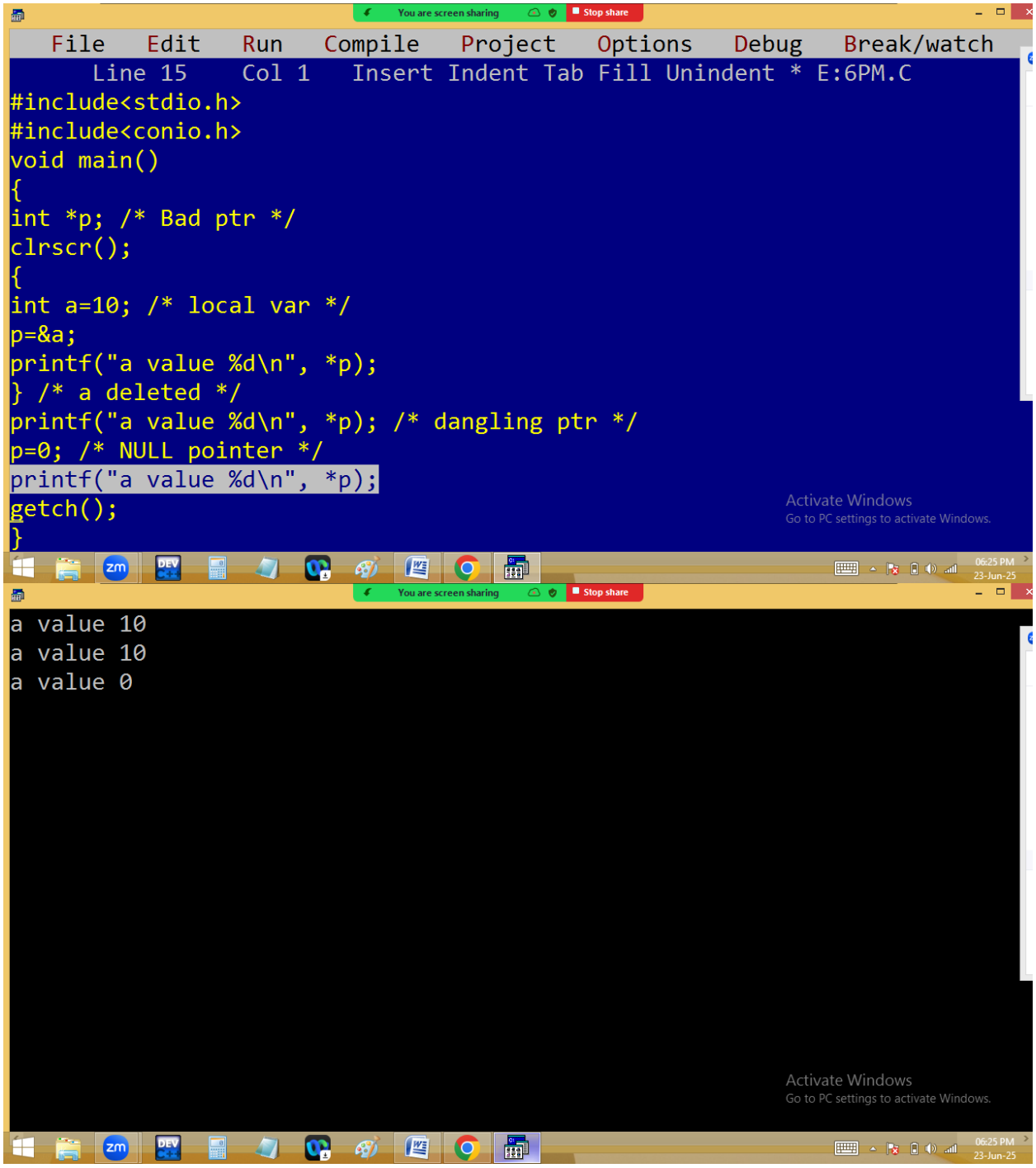
NULL pointer: When a pointer is initialized with NULL or 0 then it is a null pointer. To avoid bad and dangling pointers we are using NULL pointer.


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Stop share
File Edit Run Compile Project Options Debug Break/watch
Line 5 Col 18 Insert Indent Tab Fill Unindent * E:6PM.C
#include<stdio.h>
#include<conio.h>
void main()
{
int *p=0; /* NULL ptr */
clrscr();
printf("p stored addr %u, value %d",p, *p);
getch();
}
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06:17 PM
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Stop share
p stored addr 0, value 0
Activate Windows
Go to PC settings to activate Windows.
06:18 PM
23-Jun-25



Dangling pointer:




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Line 15 Col 1 Insert Indent Tab Fill Unindent * E:6PM.C
#include<stdio.h>
#include<conio.h>
void main()
{
int *p; /* Bad ptr */
clrscr();
{
int a=10; /* local var */
p=&a;
printf("a value %d\n", *p);
} /* a deleted */
printf("a value %d\n", *p); /* dangling ptr */
p=0; /* NULL pointer */
printf("a value %d\n", *p);
getch();
}
Activate Windows
Go to PC settings to activate Windows.
06:25 PM
23-Jun-25
You are screen sharing
Stop share
a value 10
a value 10
a value 0
Activate Windows
Go to PC settings to activate Windows.
06:25 PM
23-Jun-25

MEMORY MANAGEMENT

To store anything in our computer, we should have to allocate the memory first.

This memory allocation is conducted in two ways.

1. Static memory allocation.
2. Dynamic memory allocation.

In static memory allocation, the memory specified at compile/design time, based on the data type or array size. This type of memory management is called compile time memory management [**compiler indicates memory and O.S allocates the memory**].

In static memory allocation, the memory size is fixed at compile time and we can't

change this memory size at run time. It causes some times memory wastage / shortage.

To avoid this problem, the only solution is dynamic memory allocation.

In dynamic memory allocation, the memory is allocated at run time, based on the user input, instantly.
This type of memory management is called run time memory management.

To conduct dynamic memory allocation, we should have to use **pointers**.

In dynamic memory allocation the memory is allocated in **HEAP** area.

To manage the dynamic memory, we are using some predefined functions like

- malloc()
- calloc()
- realloc()
- free()

All these functions are available in **<alloc.h>**

malloc(), realloc(), calloc() functions are able to allocate the memory of **64KB** Maximum at a time.

To allocate more than 64KB memory, use the functions

- farmalloc()
- farcalloc()
- farrealloc().

Note:

when we are working with dynamic memory allocation, we have to allocate the

memory for any data type. Due to this all these functions return datatype is **void ***, which is a generic type. Due to this we should have to provide **explicit type casting** for all these functions.

malloc()	**calloc()**
Block Memory allocation	Contiguous memory blocks allocation
Allocates memory in bytes form	Allocates memory in blocks form.
Initial values garbage	Initial values 0
One argument required	Two arguments required
Used for normal variables	Used for array type variables

Syntax:

void * malloc(bytes);

void * calloc(no of blocks, block_size);

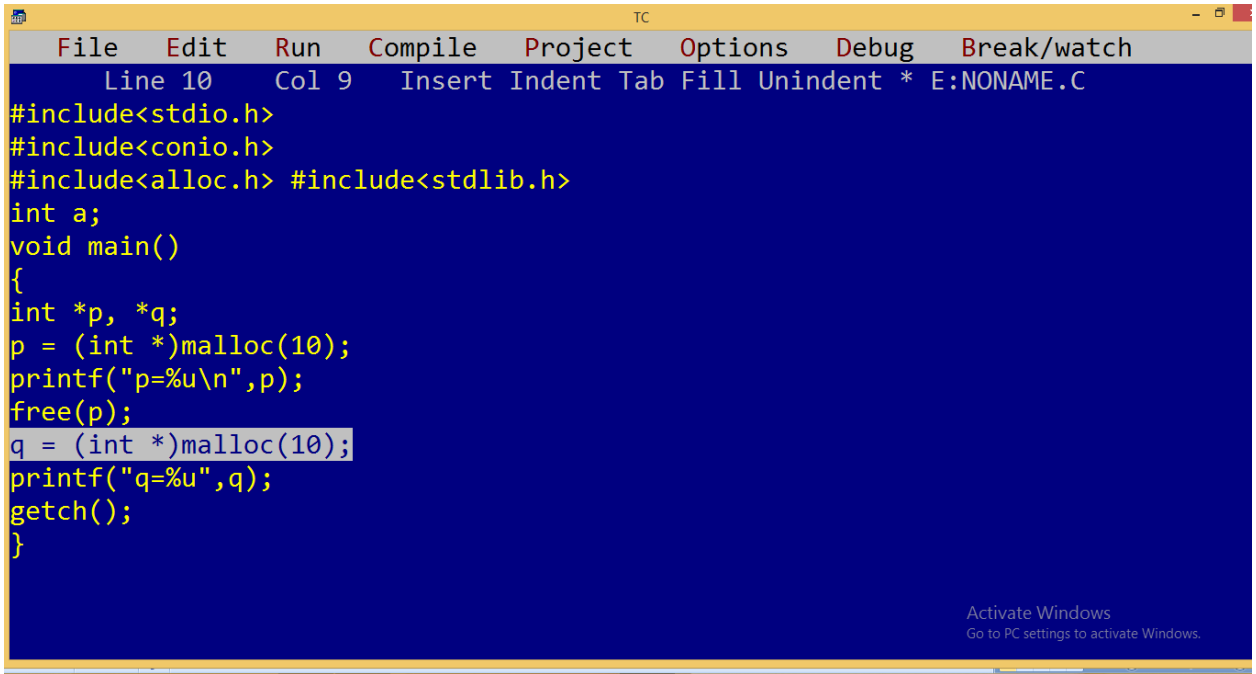
free(): It is used to release the memory allocated by malloc(), calloc() and realloc().

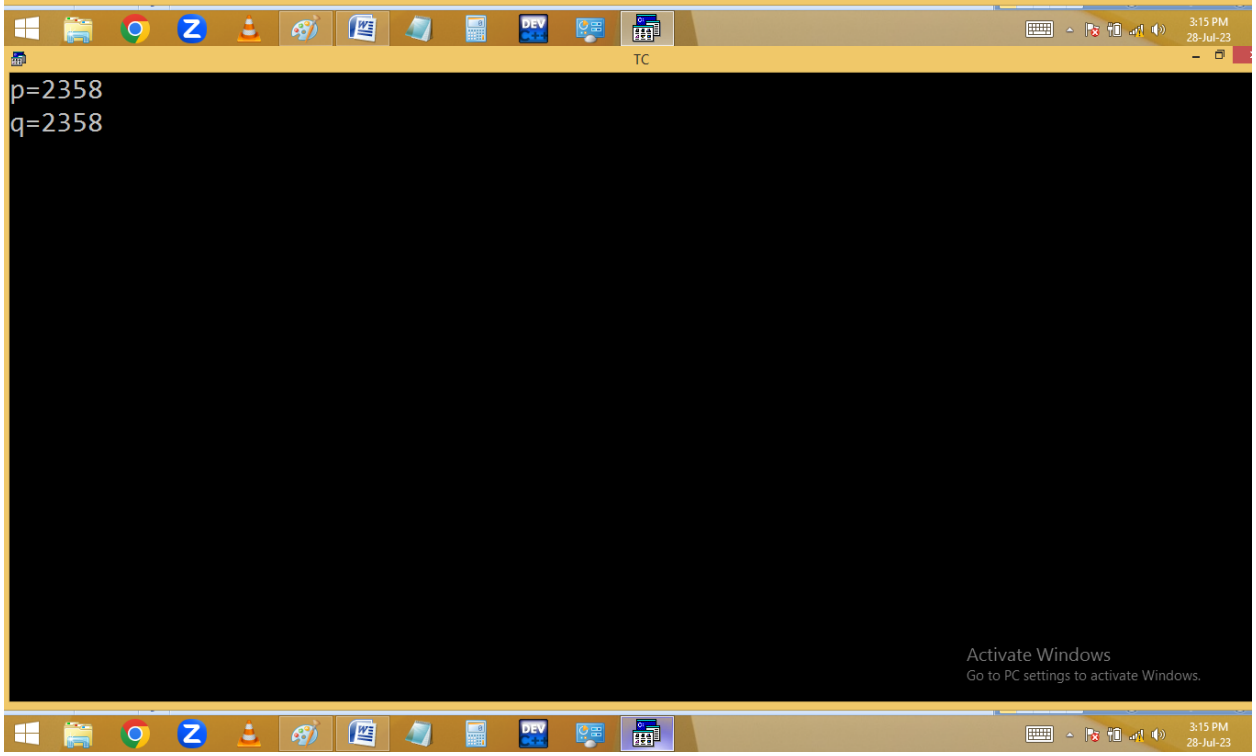
Syntax: void free(pointer);

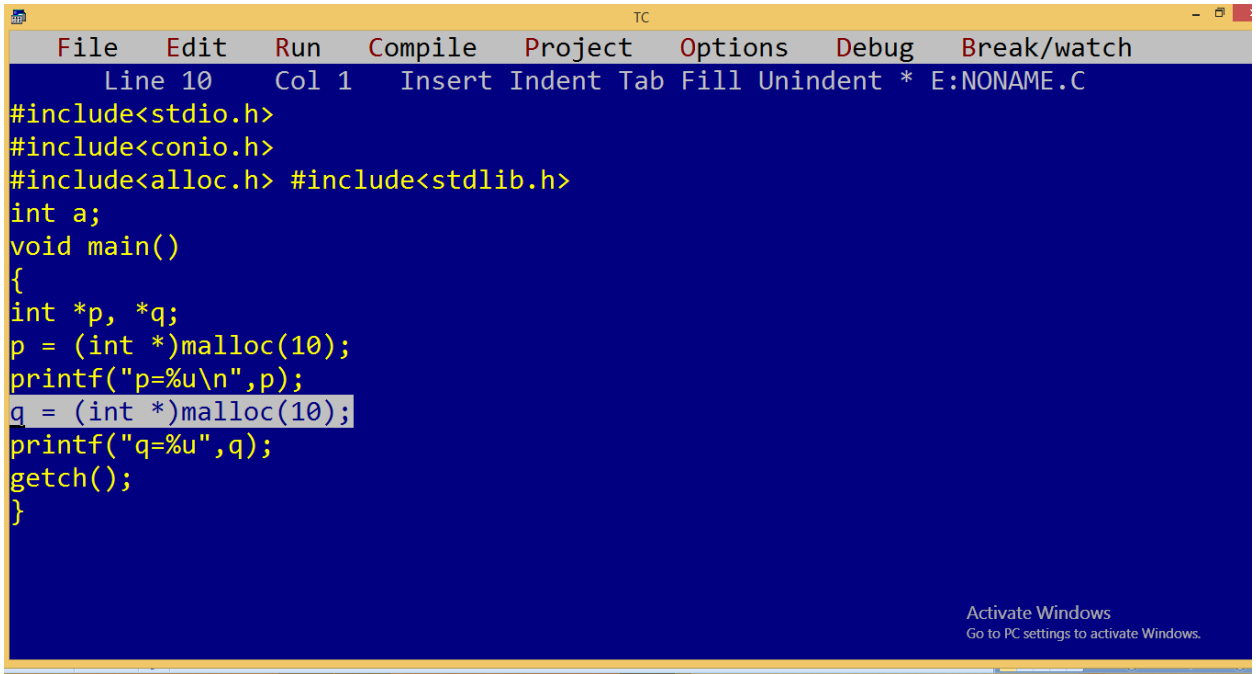
realloc(): It is used to extend the memory allocated by malloc() or calloc() at runtime. Working style is similar to malloc().

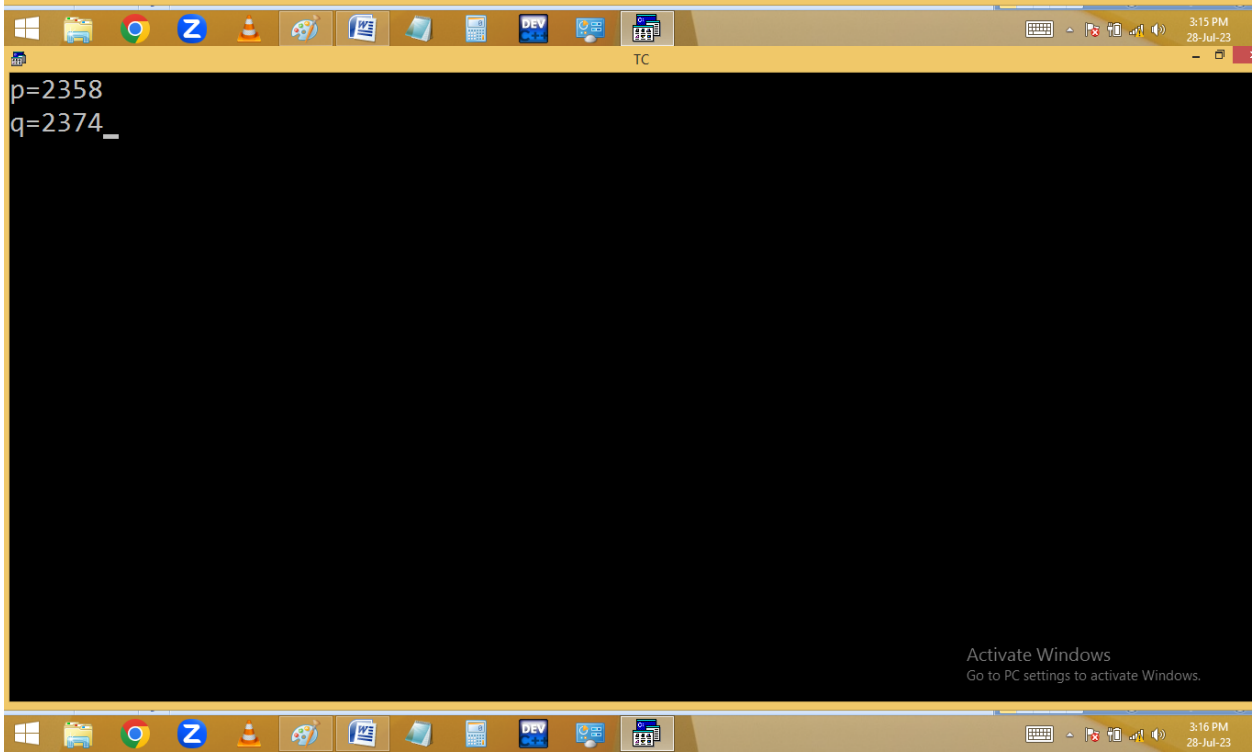
Syntax: **void * realloc(oldptr, newsize);**

free example:


TC
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Line 10 Col 9 Insert Indent Tab Fill Unindent * E:NONAME.C
#include<stdio.h>
#include<conio.h>
#include<alloc.h> #include<stdlib.h>
int a;
void main()
{
int *p, *q;
p = (int *)malloc(10);
printf("p=%u\n",p);
free(p);
q = (int *)malloc(10);
printf("q=%u",q);
getch();
}
Activate Windows
Go to PC settings to activate Windows.


TC
p=2358
q=2358
Activate Windows
Go to PC settings to activate Windows.


TC
File Edit Run Compile Project Options Debug Break/watch
Line 10 Col 1 Insert Indent Tab Fill Unindent * E:NONAME.C
#include<stdio.h>
#include<conio.h>
#include<alloc.h> #include<stdlib.h>
int a;
void main()
{
int *p, *q;
p = (int *)malloc(10);
printf("p=%u\n",p);
q = (int *)malloc(10);
printf("q=%u",q);
getch();
}
Activate Windows
Go to PC settings to activate Windows.

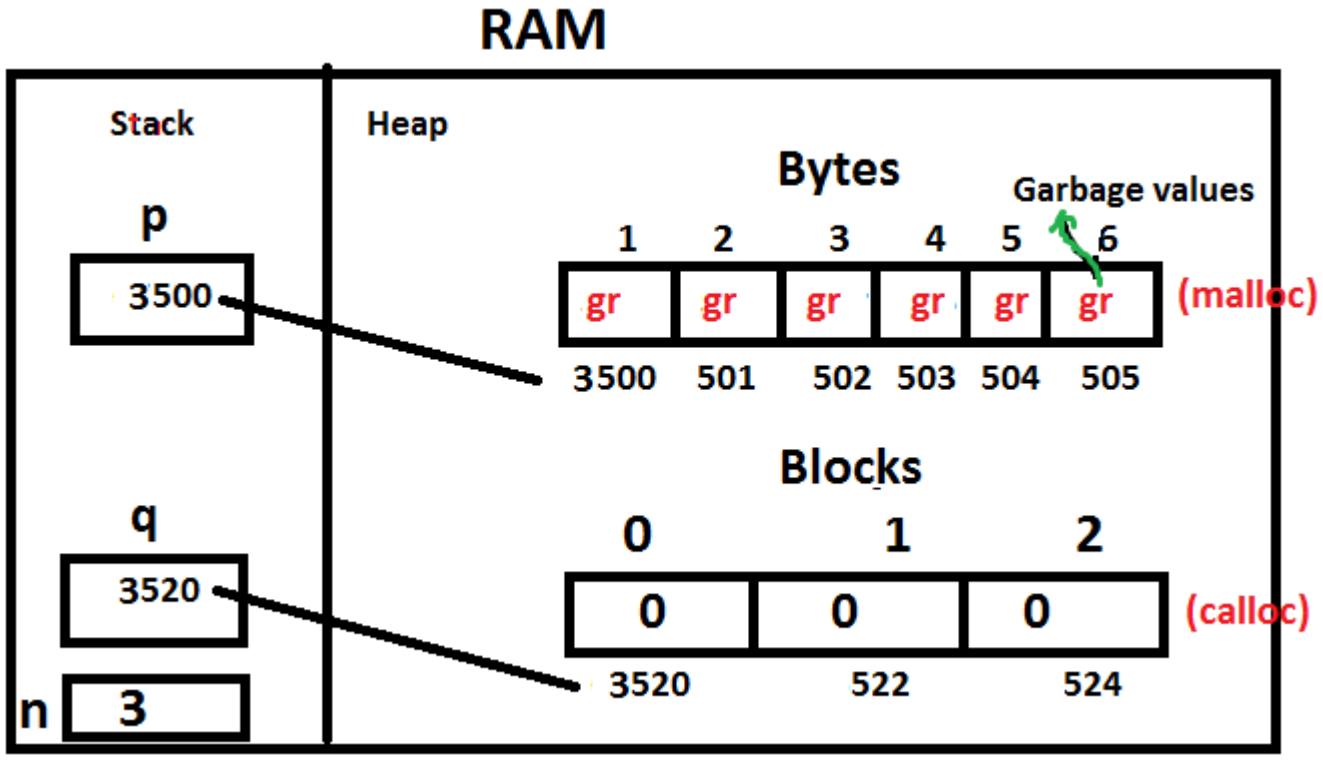

TC
p=2358
q=2374
Activate Windows
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allocating memory for 3 integers using malloc(), calloc().

```
int *p, *q, n=3;

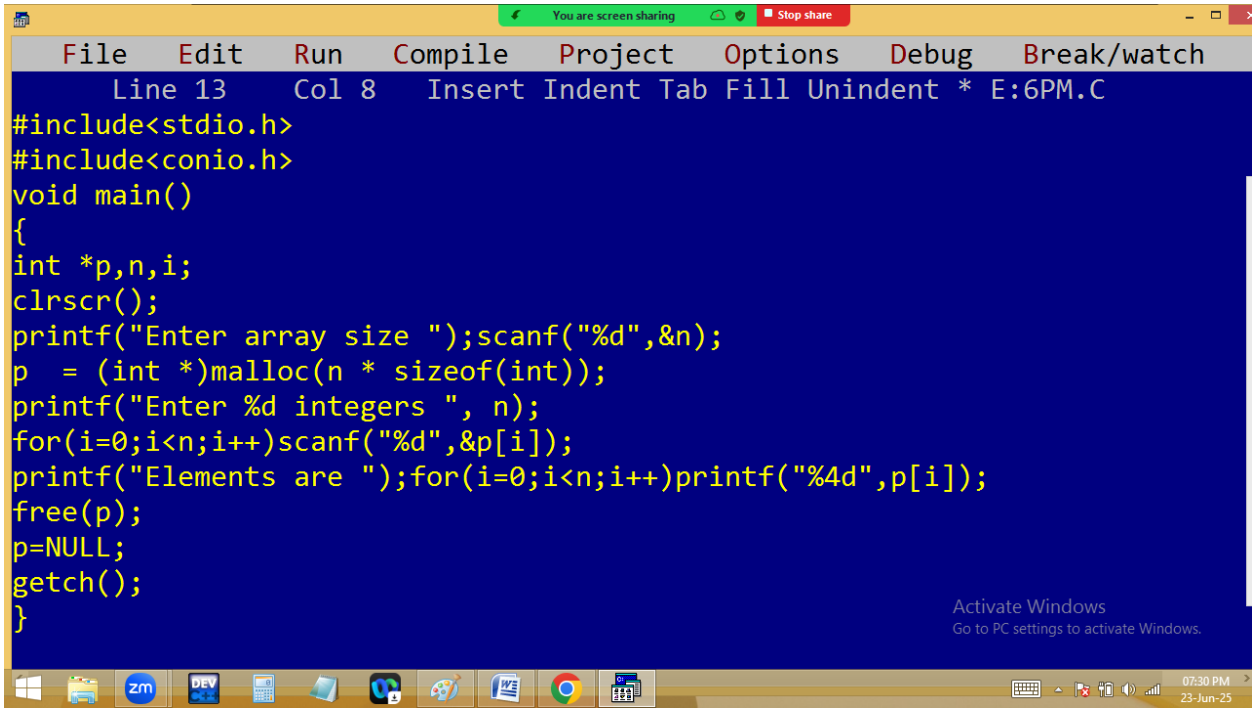
p = (int  *)malloc(n * sizeof(int));

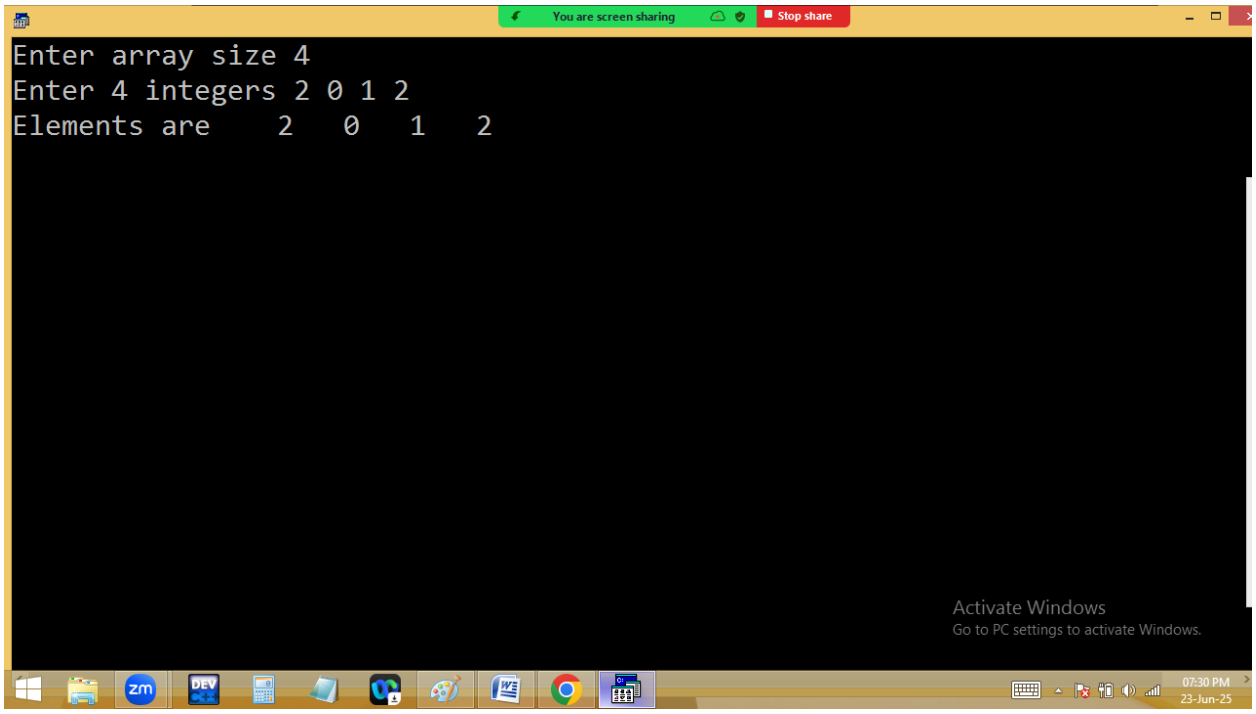
q = (int  *)calloc(n ,  sizeof(int));
```

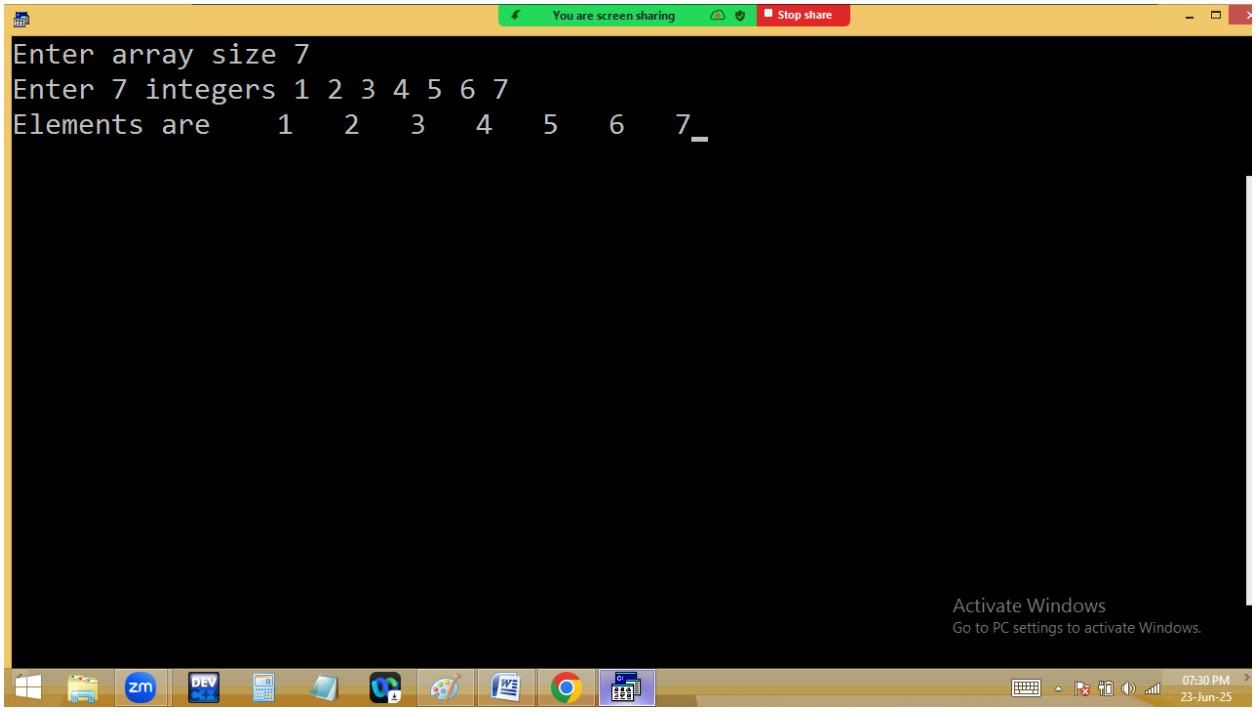


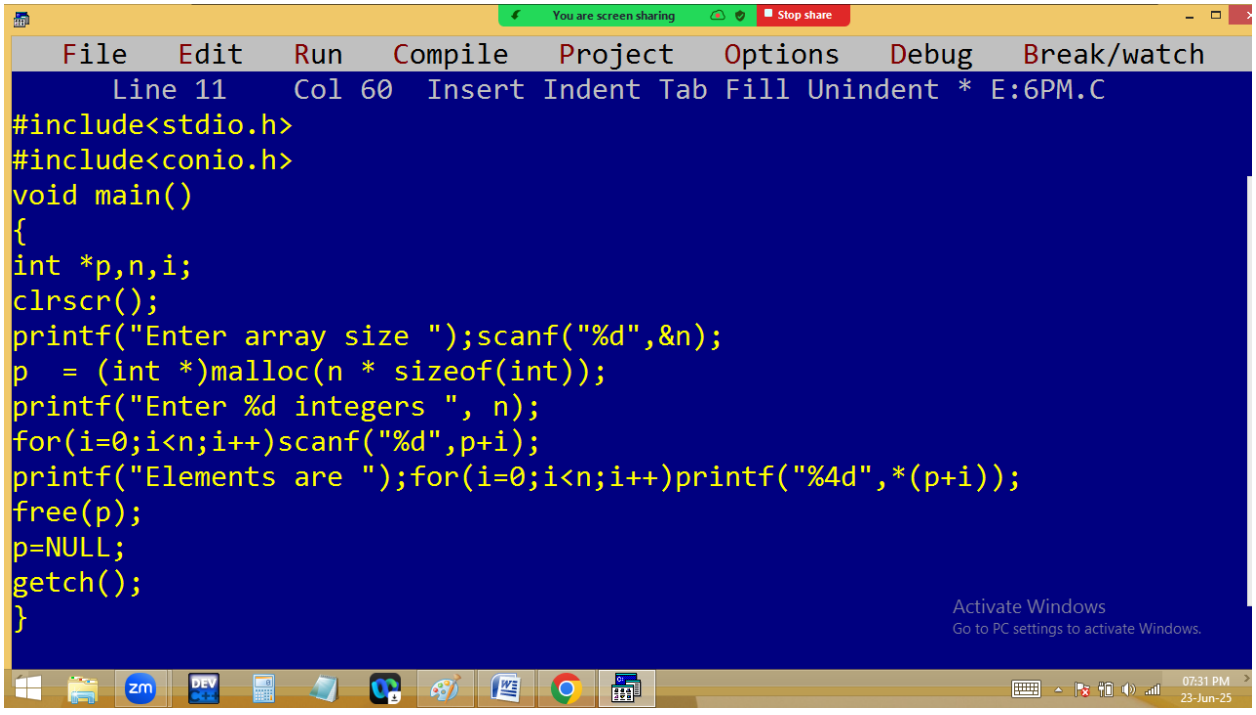
Eg:

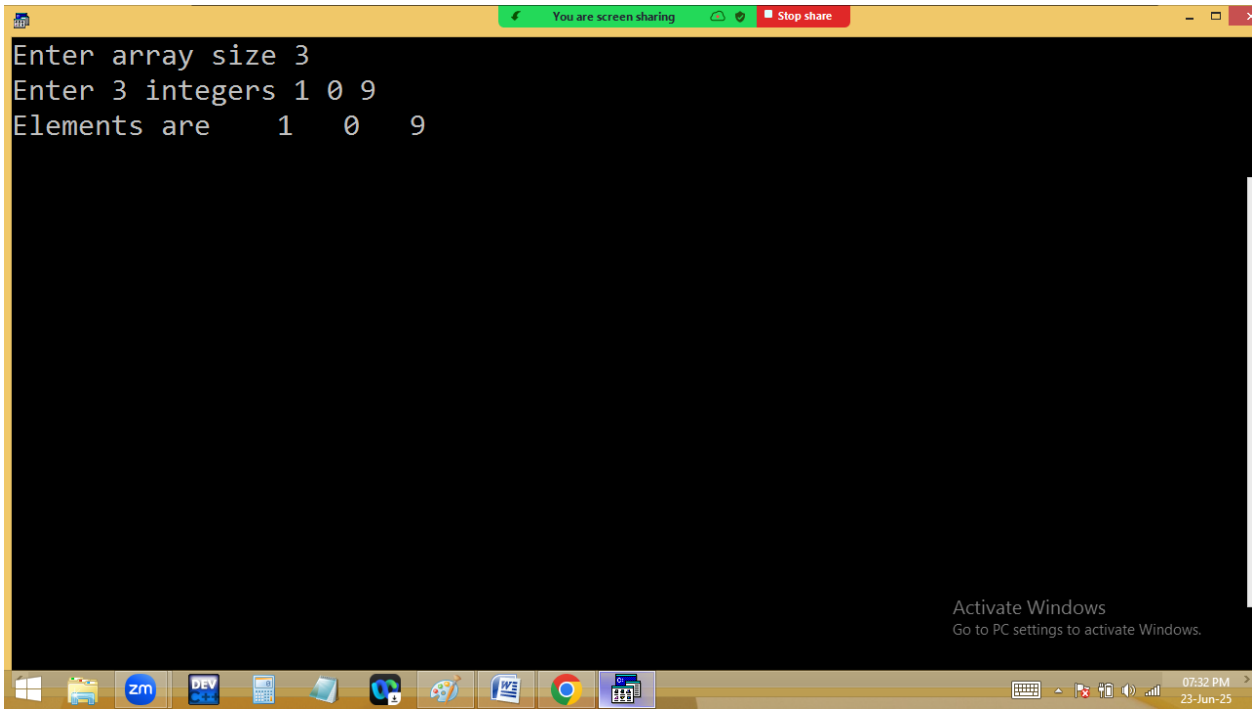
Creating dynamic one-dimensional array:

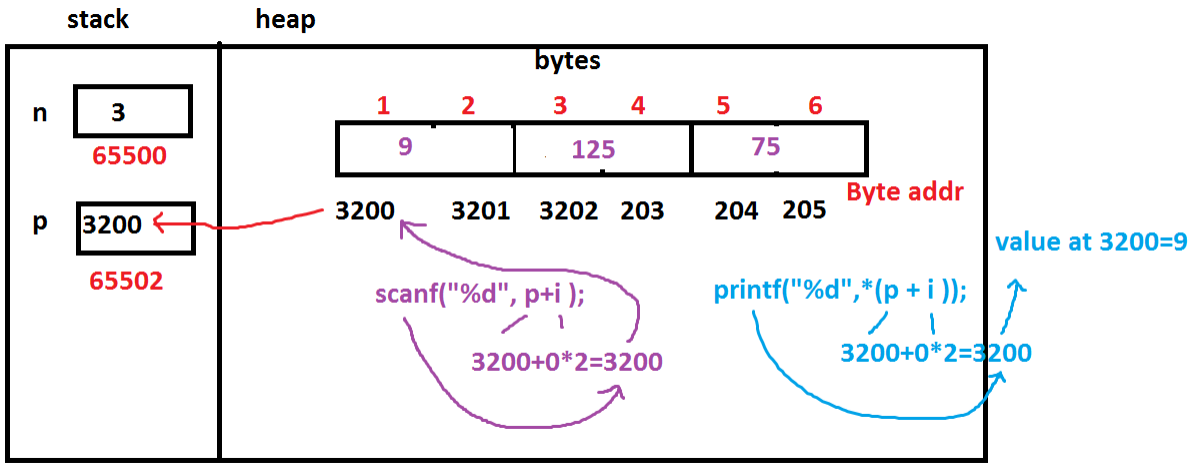

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Stop share
File Edit Run Compile Project Options Debug Break/watch
Line 13 Col 8 Insert Indent Tab Fill Unindent * E:6PM.C
#include<stdio.h>
#include<conio.h>
void main()
{
int *p,n,i;
clrscr();
printf("Enter array size ");scanf("%d",&n);
p = (int *)malloc(n * sizeof(int));
printf("Enter %d integers ", n);
for(i=0;i<n;i++)scanf("%d",&p[i]);
printf("Elements are ");for(i=0;i<n;i++)printf("%4d",p[i]);
free(p);
p=NULL;
getch();
}
Activate Windows
Go to PC settings to activate Windows.
07:30 PM
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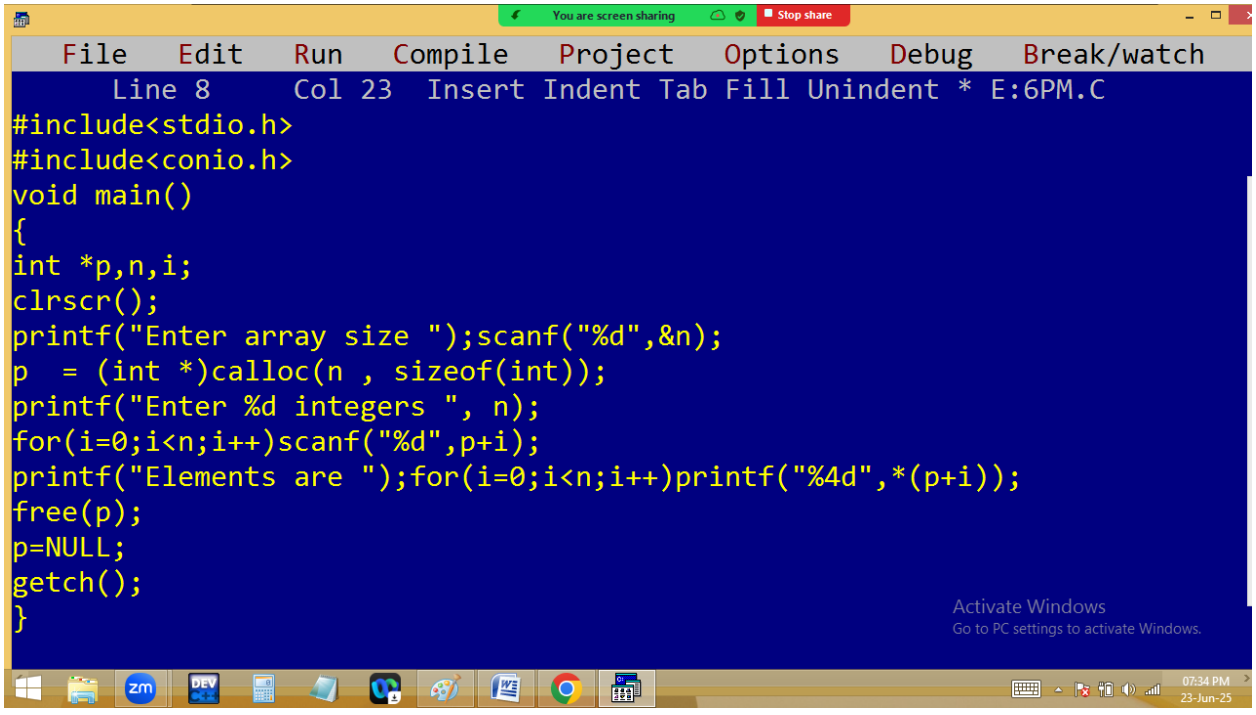

You are screen sharing
Stop share
Enter array size 4
Enter 4 integers 2 0 1 2
Elements are 2 0 1 2
Activate Windows
Go to PC settings to activate Windows.
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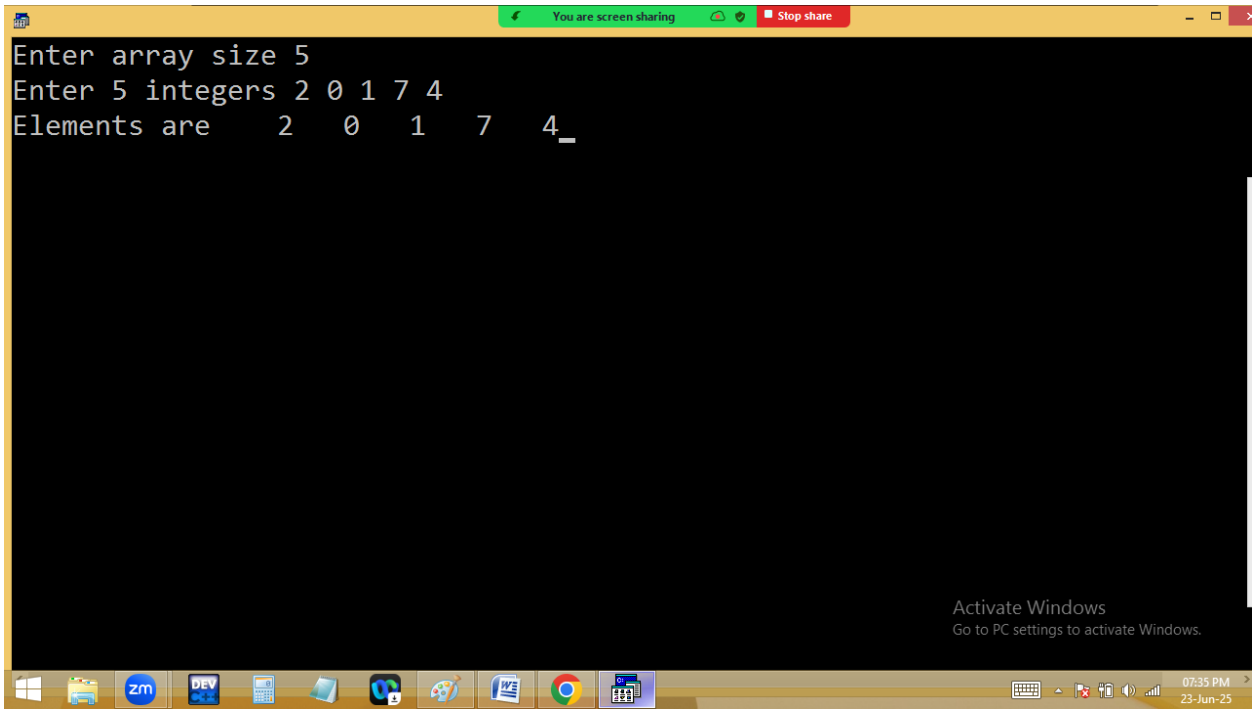

You are screen sharing
Stop share
Enter array size 7
Enter 7 integers 1 2 3 4 5 6 7
Elements are 1 2 3 4 5 6 7
Activate Windows
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You are screen sharing
Stop share
File Edit Run Compile Project Options Debug Break/watch
Line 11 Col 60 Insert Indent Tab Fill Unindent * E:6PM.C
#include<stdio.h>
#include<conio.h>
void main()
{
int *p,n,i;
clrscr();
printf("Enter array size ");scanf("%d",&n);
p = (int *)malloc(n * sizeof(int));
printf("Enter %d integers ", n);
for(i=0;i<n;i++)scanf("%d",p+i);
printf("Elements are ");for(i=0;i<n;i++)printf("%4d",*(p+i));
free(p);
p=NULL;
getch();
}
Activate Windows
Go to PC settings to activate Windows.
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You are screen sharing
Stop share
Enter array size 3
Enter 3 integers 1 0 9
Elements are 1 0 9
Activate Windows
Go to PC settings to activate Windows.
07:32 PM
23-Jun-25


stack
heap
bytes
n
3
65500
p
3200
65502
1
2
3
4
5
6
9
125
75
Byte addr
3200
3201
3202
203
204
205
scanf("%d", p+i);
3200+0*2=3200
printf("%d",*(p + i));
3200+0*2=3200
value at 3200=9


You are screen sharing
Stop share
File Edit Run Compile Project Options Debug Break/watch
Line 8 Col 23 Insert Indent Tab Fill Unindent * E:6PM.C
#include<stdio.h>
#include<conio.h>
void main()
{
int *p,n,i;
clrscr();
printf("Enter array size ");scanf("%d",&n);
p = (int *)calloc(n , sizeof(int));
printf("Enter %d integers ", n);
for(i=0;i<n;i++)scanf("%d",p+i);
printf("Elements are ");for(i=0;i<n;i++)printf("%4d",*(p+i));
free(p);
p=NULL;
getch();
}
Activate Windows
Go to PC settings to activate Windows.
07:34 PM
23-Jun-25


You are screen sharing
Stop share
Enter array size 5
Enter 5 integers 2 0 1 7 4
Elements are 2 0 1 7 4
Activate Windows
Go to PC settings to activate Windows.
07:35 PM
23-Jun-25